EDITORIAL

Juxtapapillary Uveal Melanomas

Patient Outcomes After Treatment With Proton Irradiation for Peripapillary and Parapapillary Melanomas

Treatment of uveal melanoma has favored radiotherapy with radioactive plaques or charged particle irradiation since the Collaborative Ocular Melanoma Study (COMS) showed no significant difference in survival between enucleation and brachytherapy for medium-sized tumors (COMS). The COMS melanoma-specific mortality rates were shown to be 10%, 18%, and 21% at 5, 10, and 12 years, respectively.[1] Additionally, local tumor failure rates were reported as 10.3% at 5 years. Certain eyes were excluded from the COMS Medium Tumor Trial owing to the proximity of the uveal melanoma to the optic disc. Tumors were not eligible for enrollment in the COMS Medium Tumor Trial due to proximity to the optic nerve if the tumor was (1) touching the optic disc, or (2) within 2 mm of the optic disc and broad enough to subtend an angle of 90° or greater, drawn from the center of the optic disc. These tumors present unique difficulties in treatment with radiotherapy and pose a high risk of ocular complications. In the COMS study, 9% of patients who were otherwise eligible for study inclusion were excluded owing to tumor proximity to the disc.

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In the past, juxtapapillary tumors (tumors touching or within 1 mm of the optic disc) were commonly recommended for enucleation as primary treatment. Advancements in plaque design for brachytherapy and the use of charged particle irradiation have offered promise in globe-salvaging treatment of juxtapapillary tumors once destined for enucleation. In this issue, Lane and colleagues[2] describe their unique experience with proton beam irradiation for uveal melanomas that have a tumor margin within 1 disc diameter of the optic nerve.

Charged particle irradiation, including proton-beam irradiation (PBI), delivers a homogenous dose of radiation to the tumor volume and is used as primary treatment as well as a salvage treatment for uveal melanoma. There are only a few charged particle beam facilities throughout the world that use protons and occasionally helium particles, with the group at Massachusetts Eye and Ear Infirmary being leaders in this technology. Gragoudas et al[3,4] have previously described their experience with PBI for uveal melanomas located away from the optic nerve. They reported melanoma-specific mortality rates of 15.2% and 23.5% at 5 and 10 years, respectively. These mortality rates compared favorably with other studies using charged particle radiotherapy for uveal melanoma, with a range of 12.1% to 29.7%.[5-14] Additionally, local recurrences at Massachusetts Eye and Ear Infirmary were 3.0% and 4.0% at 5 and 10 years, compared with other centers' reports of 4.0% to 9.5% and 4.3% to 7.9% at 5 and 10 years, respectively. Overall enucleation rates for Gragoudas et al[3,4] were shown to be 4.8% and 7.0% at 5 and 10 years, with other centers reporting enucleation rates of 6.1% to 19.3%. Additionally, treatment complications including vision loss occurred in most patients, depending on factors such as distance from the optic nerve and macula as well as total cumulative radiation dose.

With their demonstration of impressive results in treating uveal melanomas with PBI, Lane et al now present their findings for tumors adjacent to the optic disc (peripapillary, within 1 disc diameter of the optic nerve; juxtapapillary, within 1 mm of the optic nerve), tumors that prove to be some of the most difficult to treat with plaque radiotherapy. In this study, 573 patients with tumors within 1 disc diameter of the optic nerve were treated with PBI. Melanoma-related mortality was 11.9% and 21.3% at 5 and 10 years, respectively. Local tumor control rates at 5 and 10 years were 96.7% and 94%. Rates of enucleation were 13.3% and 17.1% (5 and 10 years), with most enucleations secondary to neovascular glaucoma. As expected, because of the close proximity to the optic nerve and macula, rates of radiation maculopathy (60.4% and 68%) and papillopathy (56.8% and 61.1%) were high at 5 and 10 years, with visual acuity loss to less than 20/200 in 79.7% and 91% of patients, respectively. This study confirms the efficacy of PBI in the treatment of uveal melanomas within 1 disc diameter from the optic nerve, with results comparable with or better than those of custom plaque brachytherapy (notched).

Notched plaque brachytherapy has also been used as a globe-salvaging treatment modality for juxtapapillary uveal melanomas. These specially designed plaques either have an indentation to allow fitting around the nerve or have removal of the plaque rim to allow lateral radiation delivery. However, other anatomical considerations must be realized to minimize plaque tilting and ensure plaque localization. Despite these limitations, several groups have shown successful tumor control with notched plaques in juxtapapillary uveal melanoma. However, long-term follow-up with notched plaques is not available and further studies are needed.

The Wills Eye Institute group described 141 patients with juxtapapillary uveal melanomas treated with plaque brachytherapy. Thirty-eight percent of tumors overhung greater than 50% of the disc, with 13% overhanging the entire disc. Notched plaques were used in 89% of patients, with 39% receiving adjuvant transpupillary thermotherapy. With a mean follow-up of 56 months, local tumor control was found to be 90%, with metastasis developing in 13% and death in 3%. Enucleation was required for 23% of patients, and final visual acuity was limited secondary to tumor location, with final visual acuity of 20/200 or worse in 77% of patients.[15] The same investigators also describe their subgroup of 37 patients with circumpapillary melanomas completely encircling the optic nerve. Notched plaques were used in 92% of patients with adjuvant treatment with transpapillary thermotherapy (49%) and argon laser photocoagulation (17%). With a median follow-up of 46 months, local tumor control was achieved in 86% of patients and metastasis occurred in 4%, with no noted melanoma-specific deaths.[16]

In addition to the use of notched plaques, plaque placement and localization are paramount to achieving success in uveal melanoma radiotherapy. Juxtapapillary tumors present significant difficulties in plaque placement secondary to posterior location as well as anatomical considerations that may cause plaque tilting or inadequate delivery of radiation to the peripapillary portion of the tumor. Using intraoperative ultrasound, posterior plaques can be placed appropriately with immediate intraoperative identification of poor localization (malposition or tilt) that may necessitate surgical plaque repositioning.[17] Plaque tilting was found to be more frequently associated with notched plaques, as well as tumors closer to the fovea and optic disc. Despite confirmation of adequate plaque placement intraoperatively, plaques may still shift postoperatively which may cause a reduction in radiation dose to the tumor apex or increased radiation dose to the optic nerve.[18] As a result, plaque placement and localization must be foremost in plaque brachytherapy, with insufficient coverage of tumor margins potentially leading to failure of local tumor control.

Using plaque placement with intraoperative ultrasound and notched plaques, Bascom Palmer Eye Institute described 31 patients with juxtapapillary uveal melanomas.[19] With a mean follow-up of 29.6 months, 100% of patients had local tumor control, whereas those treated without notched plaques had an 89% control rate at a mean follow-up of 24.7 months.[19] This and other clinical studies using notched plaques[20] with intraoperative ultrasound[17,18] to ensure optimal plaque placement demonstrate excellent local tumor control rates but continue to be associated with high rates of vision loss secondary to radiation maculopathy and optic neuropathy.

Radiation vasculopathy and optic neuropathy are the major visual threats while neovascular glaucoma is the major concern leading to enucleation in eyes with local radiotherapy tumor control with either PBI or brachytherapy. The ability to preserve the globe with uveal melanoma using radiotherapy raises the focus on managing or eliminating radiation treatment related complications. Recent study has focused on the early identification of radiation maculopathy with optical coherence tomography as well as the early treatment with anti–vascular endothelial growth factor with or without combination treatment with intravitreal corticosteroids to decrease radiation complications. Using these approaches may help stabilize visual acuity in some patients with vision loss secondary to radiation maculopathy and papillopathy.

In conclusion, the results from the COMS Medium Tumor Trial have established the widespread usage of plaque brachytherapy as a primary treatment modality for uveal melanoma. Treatment affords outcomes that are usually globe-salvaging and often spare some useful vision. Despite representing 9% of otherwise eligible tumors in the COMS study, uveal melanomas in close proximity to the optic nerve were deemed ineligible and therefore were not evaluated in this multicenter clinical trial. For these unique tumors, notched plaque brachytherapy has been shown to be efficacious, especially when performed by an experienced ocular oncologist and radiation oncologist with an emphasis on proper plaque placement and localization with intraoperative ultrasound. Charged particle radiotherapy is another primary globe-salvaging radiotherapeutic treatment that has been used with excellent efficacy. The current study by Lane et al confirms that uveal melanomas within 1 disc diameter of the optic nerve may also be treated with proton beam irradiation with outstanding long-term tumor control. Discouragingly, both PBI and brachytherapy result in significant visual morbidities secondary to radiation retinopathy and/or optic neuropathy. Future treatment strategies will need to continue to address the prevention and management of these eventual complications. Regardless of the treatment approach for uveal melanoma, tumor control must be the first and foremost focus in these patients, followed by preservation of the globe and, ultimately, vision.

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Financial Disclosure: None reported.

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Dr W. E. Lambert presented a case of lens extraction in a patient with high myopia. The young woman, aged 28 years, had been myopic since birth and in March 1910 she had myopia of 22D in each eye with only 20/200 vision with correction. There were changes in both lenses, floating opacities in the vitreous, and marked changes in the fundus of both eyes. Both were operated upon by the method of Fukala with needling and extraction. The right eye was needled in May and as the first was insufficient a second was done 3 days later with marked effect. Three days later linear extraction was performed with washing out of the cortical substance and discission later. Vision was 20/40 without correction.

Potassium iodide was given to promote absorption of the vitreous opacities.

Source: Coburn EB. Report of the Proceedings of the Section on Ophthalmology of the New York Academy of Medicine. *Arch Ophthalmol*. 1911;40:280.